Generative and Learning-Enhanced Hybrid Metaheuristics for Combinatorial Optimization

Research project for Bosnia and Herzegovina–Austria bilateral cooperation

1. Background and motivation

Combinatorial optimization (CO) is a cornerstone of modern decision support in logistics, energy systems, telecommunication networks, transportation, and data-driven planning. Graph problems such as maximum independent set (MIS), MaxCut, densest k-subgraph extraction, minimum consistent subset selection, and dynamic electric autonomous dial-a-ride problems can all be formulated as large-scale combinatorial search tasks in which the quality of the solution depends on complex structural constraints, uncertainty, and interactions between decisions. Classical exact methods provide guarantees but often become intractable on large or dynamic instances, while traditional metaheuristics can scale better but rely heavily on hand-crafted operators, parameter tuning, and domain-specific engineering (Blum and Roli, 2003; Glover and Kochenberger, 2003; Toth and Vigo, 2002).

The central challenge is not merely to find a feasible solution, but to discover high-quality, diverse, and robust search trajectories in very large combinatorial spaces. This is precisely where machine learning and generative AI can be strategically valuable. Data-driven models can identify structural patterns, approximate promising regions of the search space, and support decisions such as which vertices to include, which parts of a solution to destroy, which components to merge, and how to control the search process adaptively. Recent work in learning-enhanced combinatorial optimization has shown that machine learning can support branching, solution construction, component selection, search-space reduction, and adaptive heuristic control (Bengio et al., 2021; Mandi et al., 2024; Reixach and Blum, 2024; Reixach and Blum, 2026).

The proposed project builds on these developments and extends them by integrating generative models, including diffusion models and Generative Flow Networks (GFlowNets), into classical hybrid optimization frameworks, with particular emphasis on Large Neighborhood Search (LNS) and Construct, Merge, Solve and Adapt (CMSA). Exact subproblem optimization and, where appropriate, decomposition techniques such as column generation will complement these frameworks. The project hypothesis is that implicit knowledge encoded by graph-aware neural models and generative architectures can guide search more effectively than purely hand-designed operators, especially when combined with exact repair or incumbent-improving subproblems. This is particularly relevant when structural information matters: for example, when selecting a destroy set in LNS, identifying promising components in CMSA, or deciding when a partial solution should be re-optimized. Recent research on diffusion-based evolutionary search, GFlowNet-based destroy-set generation, and learning-enhanced CMSA provides a strong scientific foundation for this bilateral project (Akbay et al., 2025; Bresich et al., 2026a; Reixach and Blum, 2024; Reixach and Blum, 2026; Salvà Soler and Raidl, 2026; Wustinger et al., 2026).

The project is designed as a genuinely bilateral initiative in which the two partner teams contribute complementary methodological and application expertise. The Austrian partner will primarily contribute expertise in LNS, generative optimization, diffusion and GFlowNet-based search, and dynamic routing, while the Bosnia and Herzegovina partner will primarily contribute expertise in CMSA, graph-based combinatorial optimization, mathematical formulations, benchmark design, and learning-assisted component generation. The joint research will focus on transferable learning mechanisms that can be evaluated across both static graph problems and dynamic optimization settings. Two principal application domains will be considered: consistent subset selection and variants of the dynamic electric autonomous dial-a-ride problem (EADARP).

2. State of the art and research gap

The field of learning-enhanced optimization has developed rapidly. Methodological surveys show that machine learning can support branching, candidate generation, heuristic policies, algorithm configuration, and other decisions within exact and approximate solvers (Bengio et al., 2021; Mandi et al., 2024). Within metaheuristics, data-driven control has been investigated for neighborhood selection, operator scheduling, solution construction, and search-space reduction. In this context, CMSA and LNS are particularly attractive because their performance depends strongly on how promising partial solutions, components, and destroy sets are created and selected (Reixach and Blum, 2024; Reixach and Blum, 2026; Bresich et al., 2026a).

Recent advances in generative modeling provide a more expressive alternative to standard learned heuristics. Generative Flow Networks (GFlowNets) learn stochastic policies for constructing structured objects with probabilities related to their reward. Unlike purely greedy policies that aim to generate a single maximizer, GFlowNets are particularly suited to sampling diverse high-reward structures (Bengio et al., 2021; Bengio et al., 2023). This property is highly relevant to hybrid optimization. Both CMSA and LNS can benefit not only from one strong prediction, but from multiple structurally different promising candidates. GFlowNets are therefore natural tools for generating diverse candidate subsets, partial solutions, and destroy sets in graph-based optimization (Bresich et al., 2026a).

Diffusion models, originally developed for generative modeling in continuous domains, have also been adapted to combinatorial optimization. Discrete and graph-based diffusion approaches progressively transform noisy structures into high-quality candidate solutions and can be combined with classical search to improve exploration and diversification. Recent work has demonstrated the potential of diffusion-based evolutionary and adaptive search for hard graph problems, including the Maximum Independent Set and alpha-domination problems (Salvà Soler and Raidl, 2026; Wustinger et al., 2026). These methods are particularly relevant because hybrid optimization often benefits from a distribution of diverse high-quality structures rather than from a single predicted solution.

Despite these advances, important methodological questions remain open. Existing approaches typically focus on individual interactions between machine learning and optimization, such as branching, construction, neighborhood selection, destroy-set generation, or search-space reduction. The systematic integration of several learned decisions within modular hybrid metaheuristics remains comparatively underexplored. Likewise, LLM-based heuristic generation has shown promising results in automatic algorithm design, but its role inside reproducible optimization pipelines for graph and routing problems is still at an early stage.

The proposed project addresses this gap by studying a common methodological problem: **How can learning and generative models improve the construction and control of reduced search spaces inside hybrid metaheuristics?**

The project will investigate this question at three complementary levels: 1) **offline learning** from previous optimization experience; 2) **online adaptation** from the current search trajectory; 3) **generative construction and automatic heuristic discovery.** Rather than attempting to develop a universal neural optimizer, the project aims to identify principles that can be transferred across different optimization frameworks and problem classes.

3. Scientific objectives

The project pursues the following objectives:

(i) To design graph-aware learning mechanisms for component selection, destroy-set generation, and constructive heuristics in hybrid optimization methods such as CMSA and LNS.

(ii) To integrate diffusion models, GFlowNets, and related generative approaches into hybrid search procedures for graph optimization, with emphasis on diversity, quality, and multi-objective performance.

(iii) To develop decision-focused learning strategies for dynamic routing and scheduling applications, especially in the context of EADARP and related online decision problems.

(iv) To investigate the use of large language models (LLMs) as generators of lower-level heuristics, scoring functions, and search-control rules that can be tested and integrated within a reproducible algorithmic framework.

(v) To validate these methods on benchmark graph problems and applied decision scenarios including the minimum consistent subset problem, maximum independent set, MaxCut, dense k-subgraph problems, and dynamic EADARP.

The overall goal is to create a hybrid algorithmic pipeline that fuses the robustness of exact subroutines and metaheuristics with the expressiveness and generalization power of modern generative and neural techniques.

4. Proposed methodology and work packages

The project will be organized around a set of interconnected work packages that mirror the repository structure and the scientific agenda of the proposal.

WP1. Modular hybrid optimization framework

This work package will establish a modular optimization framework centered on CMSA- and LNS-based search. Constructive processes, component generation, destroy/repair mechanisms, and reduced-subproblem optimization will be exposed through common interfaces so that learned policies and generative models can be evaluated under comparable conditions. Exact subsolvers will be used where appropriate, while decomposition mechanisms such as column generation may be incorporated for problem classes in which they provide a natural advantage. The aim is not to force all techniques into a single monolithic algorithm, but to create a common experimental backbone for studying how learning can improve the generation and control of reduced search spaces (Akbay et al., 2025; Reixach and Blum, 2024; Reixach and Blum, 2026).

WP2. Diffusion models and GFlowNets for destroy-set and candidate generation

The second work package will focus on generative architectures for combinatorial structures. Diffusion models and GFlowNets will be used to generate structured destroy sets, candidate subsets, and component pools that are more informative than randomly sampled neighborhoods. For LNS, the learned model will propose destroy sets that target critical vertices, edges, or components capable of triggering strong repair phases. For CMSA, the model will generate candidate components or partial solutions whose inclusion is likely to improve the objective value. The key research challenge is to combine diversity, feasibility, and reward-aware generation in a way that is compatible with exact repair and search intensification. This direction is aligned with recent work on diffusion-based search and GFlowNet-guided neighborhood generation (Bengio et al., 2021; Bengio et al., 2023; Bresich et al., 2026a; Salvà Soler and Raidl, 2026; Wustinger et al., 2026).

WP3. Generative AI (LLMs) for lower-level heuristic synthesis

This work package will explore large language models as offline algorithm-design assistants. LLMs will be used to propose candidate lower-level heuristics, repair operators, scoring functions, neighborhood definitions, and search-control rules from a high-level specification. Generated routines will be translated into executable code, automatically filtered by feasibility and safety checks, and evaluated using objective optimization performance. The goal is not to replace the solver or to invoke an LLM at every search iteration, but to create a reproducible human-AI co-design framework in which LLMs propose candidate algorithmic ingredients and the optimization framework selects them according to empirical evidence. This exploratory direction is motivated by recent work on LLM-based algorithm discovery and heuristic evolution (Romera-Paredes et al., 2024; Liu et al., 2024; Yang et al., 2023).

WP4. Application 1: Minimum consistent subset problem

The first major application domain will consist of graph-based combinatorial optimization problems, with particular emphasis on the **Minimum Consistent Subset (MCS)** problem.

Let $G=(V,E)$ be a connected graph in which each vertex has one of k colors. A subset $S \subseteq V$ is consistent if, for every vertex $v \in V$, the vertices of S that are closest to v contain at least one vertex having the same color as v. The Minimum Consistent Subset problem asks for a consistent subset of minimum cardinality.

This problem provides a natural model of **representative or prototype selection on colored graphs**. Vertices may represent objects, observations, or agents; graph distances describe relationships or similarity; and colors represent classes or categories. Application contexts include graph-based classification, representative selection, network summarization, and networked data analysis. MCS is especially suitable for learning-enhanced optimization because the relevance of a vertex depends simultaneously on its color, its structural position in the network, shortest-path relations, and the current set of selected representatives.

The project will benchmark exact formulations, CMSA- and LNS-based hybrid methods, and learned component-generation strategies on synthetic and real-world instances, and compare them with relevant state-of-the-art approaches from the literature.

WP5. Application 2: (Dynamic) electric autonomous dial-a-ride problem

The second application is the dynamic EADARP. This problem includes pickup and delivery requests, fleet constraints, charging-station availability, time windows, battery limitations, and dynamically arriving demand. It provides a realistic setting in which learning-based anticipatory decisions may be particularly valuable. The project will use static EADARP instances to calibrate the hybrid framework and then extend it to dynamic variants in which the solver must decide when to re-optimize, which requests or routes to reconsider, and how to adapt charging decisions online. The two principal application domains deliberately represent complementary settings: MCS is a static graph-based selection problem, whereas EADARP is a dynamic routing problem. Their joint study will therefore test whether the proposed learning mechanisms transfer across substantially different optimization environments (Bresich et al., 2026b; Tomandl et al., 2026; Bresich et al., 2026c).

Experimental evaluation

The proposed learning-enhanced methods will be evaluated against conventional CMSA and LNS baselines, manually designed operators, exact approaches where computationally feasible, and relevant recent methods from the literature. Performance will be assessed using objective value or optimality gap, runtime, time-to-target, anytime performance, robustness over repeated runs,

diversity of generated candidates, reduced-subproblem size, and generalization to larger or structurally different instances. Training, inference, and optimization times will be reported separately in order to assess whether the computational cost of learning is justified and amortized across repeated problem solving.

Risk and contingency

The modular design limits the scientific risk of individual learning components. If GFlowNet- or diffusion-based generation does not outperform conventional construction, the same framework will support simpler neural or graph-based scoring models. If online neural adaptation proves unstable or too expensive, lightweight contextual-bandit or adaptive-probability mechanisms will be used. LLM-based heuristic synthesis is explicitly exploratory and is not a critical dependency for successful project completion.

5. Expected results and impact

The project will generate several tangible outputs. First, it will produce a modular hybrid optimization framework that combines classical search with graph-aware learning, generative models, and decision-focused control. Second, it will deliver application scenarios covering MCS, Maximum Independent Set, MaxCut, densest k-subgraph, and dynamic EADARP variants. Third, it will create reusable algorithmic modules for destroy-set generation, candidate construction, and adaptive heuristic control. Fourth, it will produce open and reproducible software components aligned with the planned work packages.

The expected scientific impact lies in demonstrating how generative AI and machine learning can be embedded within robust optimization algorithms rather than used only as isolated end-to-end models. The project will generate methodological knowledge on when learned search components provide measurable benefits, how well they generalize across instance distributions, and how their computational cost should be balanced against optimization gains. The bilateral collaboration is also expected to produce joint publications, thesis topics, researcher mobility, and reusable software with potential relevance to logistics, network analysis, infrastructure planning, and large-scale graph decision making.

6. Bilateral cooperation and division of responsibilities

The bilateral dimension is integral to the scientific design rather than an administrative add-on. The project deliberately couples two complementary algorithmic traditions and two distinct application settings. The Austrian partner will lead work on LNS, generative optimization, diffusion and GFlowNet-based search, and dynamic routing/EADARP. The Bosnia and Herzegovina partner will lead work on CMSA, graph combinatorial optimization, mathematical formulations, MCS, benchmark construction, and learning-assisted component generation.

The partners will jointly design the common interfaces between learning and optimization, compare offline and online learning strategies, study cross-problem and out-of-distribution generalization, and develop the LLM-assisted heuristic-synthesis pipeline. Research visits will be organized around concrete joint tasks—algorithm integration, model training, benchmark generation, computational experimentation, and paper preparation—so that mobility directly supports scientific deliverables.

This division creates clear scientific added value. The Bosnia and Herzegovina team contributes graph-oriented modeling and CMSA-based reduced-subproblem expertise, while the Austrian team contributes LNS, generative learning, and dynamic optimization expertise. The common methodology will be tested across both teams' problem domains, enabling a stronger assessment of transferability than either partner could obtain by working only within its established application area. The collaboration will also support co-supervision of graduate students and early-stage researchers, shared software development, joint publications, and preparation of follow-up European proposals at the intersection of OR and AI.

7. References

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